23CSE111 OBJECT ORIENTED PROGRAMMING
S2 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus
CASE STUDY
PHASE – I (Analysis)
Problem Identification & Class Mapping (CO1)

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1. Problem Title : Attendance Management System

2. Problem Description :

Manual attendance tracking in educational institutions is time-consuming and prone to errors. Teachers usually record attendance in registers, which makes it difficult to monitor attendance percentage and identify students with low attendance. Students also cannot easily track their attendance status.

An Attendance Management System automates the process of recording, storing, and monitoring attendance digitally. It allows teachers to mark attendance efficiently and enables students to view their attendance records and percentages. The system improves transparency, reduces manual work, and helps institutions maintain accurate attendance records.

3. Identified Entities (Classes)

Class Name	Description
Student	Represents a student who can view attendance records and percentages
Teacher	Represents a teacher who marks and manages attendance
Attendance	Stores attendance records of students for each subject
Course/Subject	Represents the subject for which attendance is recorded
Admin	Manages system accounts and generates reports

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
Student	studentID, name, course	viewAttendance(), checkAttendancePercentage()
Teacher	teacherID, name, subject	markAttendance(), editAttendance()
Attendance	date, studentID, subject, status	recordAttendance(), updateAttendance()
Admin	adminID, username, password	manageAttendance(), generateReports()